

João Roberto Lawall Soares

Founding Engineer | DeFi & DEX Architecture | Smart Contracts (Solidity, Rust)

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Summary

Engineer building decentralized exchanges and DeFi infrastructure across the smart contract layer, off-chain indexers, keepers, and wallet integrations. Shipped a ZK dark-pool DEX with batch auctions that reduces MEV exposure, a Solana stablecoin compliance protocol verifying Groth16 proofs in under 200K compute units, and production smart contracts in Solidity (Ethereum) and Rust (Solana). 6+ years building production Go systems with tight security and reliability budgets. Used to making pragmatic calls in early-stage environments and mentoring the engineers around me.

Core Skills

Smart Contracts: Solidity, Rust (Solana), Anchor, Token-2022, Foundry, Hardhat, Remix IDE, reentrancy guards, gas optimization

DeFi & DEX: AMMs, order books, batch auctions, hybrid models, liquidity mechanisms, settlement, MEV mitigation, front-running protection, DEX architecture

Blockchain Infrastructure: Ethereum (EVM), Solana, NEAR Protocol, oracles, bridges, cross-chain, SPL tokens, MetaMask, Phantom Wallet integrations

Zero-Knowledge: halo2, arkworks, Groth16 (BN254), Noir, ZK proofs, proof aggregation, on-chain verifiers, Light Protocol state compression

Off-Chain Infrastructure: Indexers (Helius webhooks), keepers, relayers, REST APIs (Go, axum), event-driven architecture, Apache Kafka, Arweave

Backend & Frontend: Go, Rust, TypeScript, Node.js, NestJS, Next.js, React, Angular, WASM, PostgreSQL, Redis

Security & Operations: Smart contract testing, security reviews, monitoring, Docker, Kubernetes, CI/CD pipelines, Linux

Leadership: Mentored 60+ Web3 engineers, led 5-engineer team, ran 20+ Solidity workshops

Professional Experience

Borderless Coding

London Area, UK (Remote)

Web3 Technical Mentor & Ambassador

Sep 2025 – Present

Part-time

- Mentored 60+ engineers on Solidity smart contract development, DEX architecture, and DeFi protocol design with a 95% satisfaction rate across cohorts.
- Ran 20+ live workshops on smart contract development, security patterns, and gas optimization using Foundry, Hardhat, and Remix IDE. 90% completion rate.
- Wrote hands-on material covering Ethereum, Solana, AMMs, order books, oracles, and cross-chain primitives. Community grew 200%.

Platform Builders

Brazil (Remote)

Senior Backend Engineer

Aug 2025 – Present

Full-time | RD Saúde – Brazil's largest pharmaceutical retailer

- Fixed a concurrency bug in a distributed pipeline processing 140M+ monthly transactions. Applied mutex-based synchronization to eliminate the race condition with zero downtime.
- Added mutex-based locking to Apache Kafka consumers, reaching 99.99% idempotency and stopping duplicate event processing across the microservice cluster.
- Migrated database layer from Oracle to PostgreSQL. Rewrote execution plans and added targeted indexes, cutting query times by over 1 minute on the slowest queries.
- Shipped 15+ features in Go with zero production incidents. Reworked goroutine management to reduce system latency by 40%.
- Mentored 5 engineers on Go concurrency patterns and distributed system architecture. Team velocity went up 30% over the following quarter.
- Tech Stack: Go, Apache Kafka, PostgreSQL, Docker, Kubernetes, CI/CD pipelines, REST APIs, microservices.

Docile Alimentos LTDA

Lajeado, RS, Brazil (Remote)

Backend Engineer

Jan 2020 – Aug 2025

Full-time

- Built Go backend services with concurrency patterns that raised throughput by 50% in production systems with tight reliability and security requirements.
- Broke apart a monolith into microservices behind REST APIs. Performance improved 40%.
- Rewrote indexes on 20+ PostgreSQL queries (45% average speedup) and reworked MongoDB aggregation pipelines, cutting query times by 85% (30s reduction).
- Shipped 50+ features and fixes across the stack on schedule (98% on-time delivery).
- Tech Stack: Go, Node.js, React, Angular, Docker, Linux, PostgreSQL, MongoDB, REST APIs.

DeFi & Web3 Projects

DarkPool-Exchange | ZK Dark Pool DEX with Batch Auctions • [GitHub](#)

Hybrid DEX where orders stay private until settlement. Traders encrypt orders to the operator and the engine matches them in periodic batch auctions (5s rounds) with a uniform clearing price per round, cutting front-running and MEV exposure. Built the Rust and on-chain layer end to end: halo2 / arkworks circuits proving order validity, client-side proof generation in Rust (WASM and native CLI), a proof aggregator

that batches per-order proofs into a single on-chain check for gas efficiency, and integration with the Solidity verifier and settlement contract. Order size and price surface only after the round closes.

zkSettle | ZK Compliance Infrastructure for Solana Stablecoins • [GitHub](#)

Solana compliance layer for stablecoin protocols. Proves travel-rule and sanctions conformance without putting user PII on-chain. Built the Rust and on-chain side: Anchor program with a Token-2022 Transfer Hook that verifies Groth16 proofs (BN254) via Solana `alt_bn128` syscalls in under 200K compute units, nullifier storage on Light Protocol state compression, plus the off-chain Rust workspace (issuer service, OFAC sanctions updater with a Sparse Merkle Tree, Helius-webhook indexer pinning attestations to Arweave, axum API gateway). Browser-side proof generation via Noir + WASM keeps PII off the device.

Crypto-Tip | Cross-Chain Tipping Smart Contracts • [GitHub \(34 Stars\)](#)

Smart contracts in Solidity (Ethereum) and Rust (Solana) for open-source funding. Wallet integrations with MetaMask and Phantom. Reentrancy guards, access control, and event-driven on-chain accounting.

givemoney.fun | Web3 Streaming Donation Platform

Solidity (Ethereum) and Rust (Solana) smart contracts for streamers to receive on-chain donations. Built reentrancy guards, withdrawal queues, and wallet integrations.

Pattpay | On-Chain Subscription Payments on Solana • [GitHub \(Cypherpunk Hackathon\)](#)

Recurring billing on Solana with sub-second finality. Smart contracts handle authorization, scheduled SPL token transfers, and on-chain accounting under tight compute budgets.

Proof Of Learn | NEAR Protocol + ZK Credentials • [GitHub \(NearX & ZKVerify Hackathon\)](#)

NEAR Protocol smart contracts that issue NFT credentials backed by zero-knowledge proofs. Integrates AI-generated learning roadmaps with on-chain incentive mechanisms.

Open Source Contributions

Zed Editor • [GitHub \(70k+ Stars\)](#) – Rust

56 PRs (30 merged) across 15+ subsystems, totaling 6,800+ lines in Rust:

- Fixed a macOS deadlock in the GPUI rendering framework where re-entrant key status changes froze the UI. ([#51035](#) – Merged)
- Reworked picker selection logic across 11 files so non-selectable entries no longer trigger unintended actions. ([#50705](#) – Merged, +185/-51)
- Fixed visible escape characters in LSP diagnostic messages rendered via markdown. ([#51766](#) – Merged, +195/-27)
- Fixed semantic tokens not appearing when opening a buffer from multibuffer. ([#53712](#) – Merged)
- Additional contributions to GPUI, DAP debugger, theme selector, markdown rendering, terminal, and platform-specific macOS/Linux fixes. ([View all PRs](#))

Education

Universidade do Vale do Taquari – Univates

2021 – 2025

Bachelor’s Degree, Computer Science

Languages

English – Professional Working Proficiency **Portuguese** – Native Speaker